Universidad Internacional del Ecuador (UIDE)

Computer Science School

Network Communication Technology

Title: *Phase 2: Packet Tracer Network Prototype*

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Summary

This report presents the Phase 2 prototype of the network for the Chat Empresarial project. The design of Phase 1 was built in Cisco Packet Tracer 9.0.1 with two sites joined by a routed link. Site A has the users and the servers (chat and authentication) and Site B has users. The prototype uses the same addressing plan as Phase 1 and was tested with real traffic. The test results and the device configurations are backed by screenshots taken from the working simulation. Two items, the lease table of Table 4.3 and Problem 3 in section 6.2, are documented from the console output only.

The test matrix has 15 tests: 14 passed and 1 failed. The failed test (T-07) was the first ping between the two sites. It failed because the routers had no route to the remote site. After the static routes were added, the same test was repeated as T-07b and passed (section 6.2).

Where to find each required item

Required content	Where
Physical topology diagram	Section 1, Figure 1
Logical topology diagram and changes from Phase 1	Section 1, Figure 3 and Table 1.1
Addressing table with VLSM	Section 2, Table 2.1
Device and interface table	Section 3, Table 3.1
Services: DHCP, static routing, SSH	Sections 4, 5 and 6
Configurations saved	Section 10.4
Configuration backups	Section 10, Table 10.2 (files in the repository)
Preliminary test matrix	Section 7, Table 7.1
Problems found and how they were solved	Section 6.2
Remaining work and requirements status	Section 8
Updated schedule through Week 11	Section 9, Table 9.1
Contributions, AI use and resources	Section 10
Repository link	Section 10.4

1. Topology and changes from Phase 1

The prototype keeps the two-site design of Phase 1. Site A holds the users (10.10.0.0/26) and the servers (10.10.0.128/27) behind router R1. Site B holds the users (10.10.0.64/26) behind router R2. R1 and R2 are joined by the link 192.168.100.0/24. Each site has its own switches and each end device has a fixed port. Figure 1 shows the cabling with all ports and addresses, and Figure 2 shows the same network in the Packet Tracer workspace.

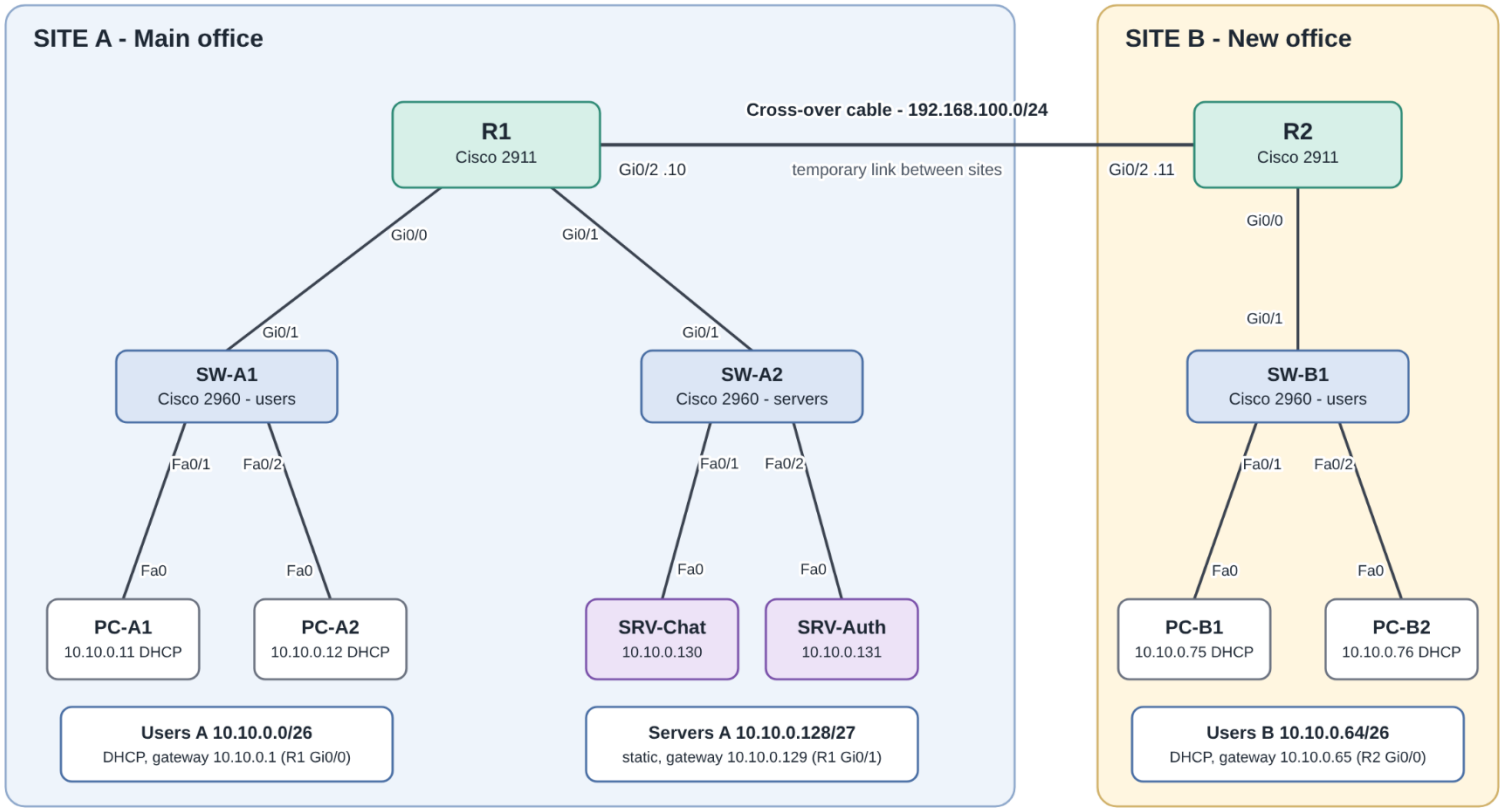


Figure 1. *Physical topology of the prototype with ports, addresses and subnets.*

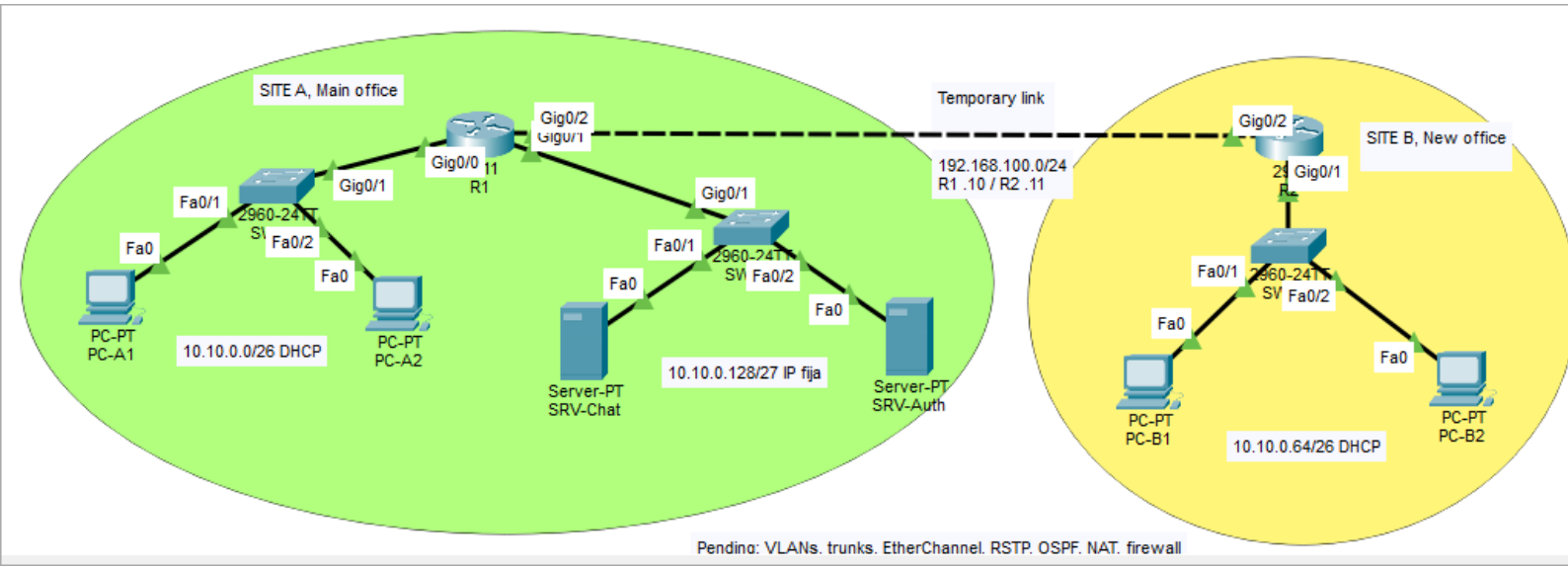


Figure 2. *The built network in the Packet Tracer 9.0.1 workspace.*

Figure 3 is the updated logical architecture. Solid boxes are what exists in the prototype. Dashed boxes are what is planned for the next weeks.

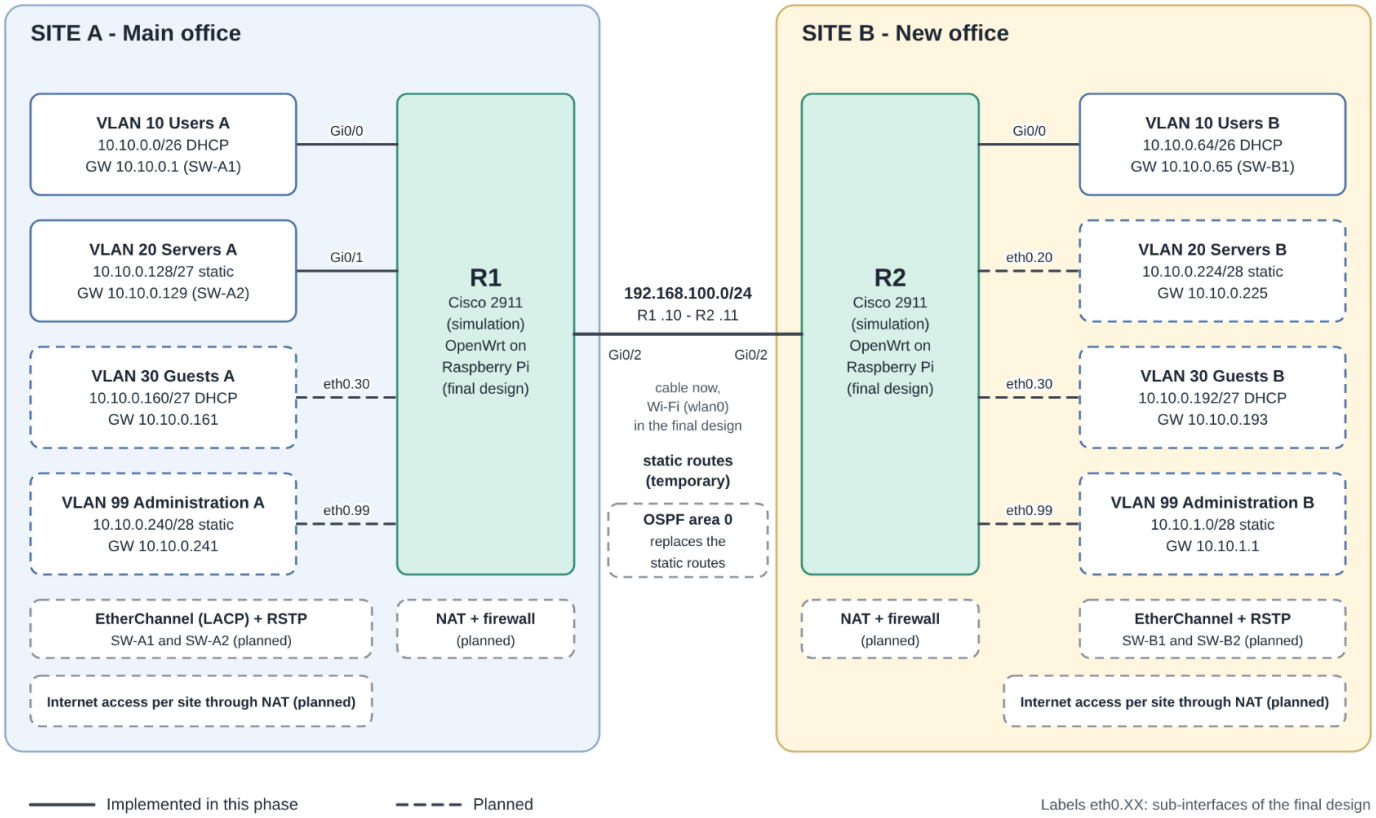


Figure 3. *Updated logical architecture. Solid: implemented. Dashed: planned. The VLANs are not configured yet: in the prototype all switches are in VLAN 1, and the VLAN names show the planned design.*

1.1 Changes from Phase 1 and why

Topic	Phase 1 design	Phase 2 prototype	Reason
Routers	OpenWrt on Raspberry Pi	Cisco 2911 routers (R1, R2)	Packet Tracer has no OpenWrt device. The routing, DHCP and SSH functions are the same and are mapped to OpenWrt in section 8.
Link between sites	Wi-Fi between the two Raspberry Pi	Wired cross-over cable on Gi0/2 of each router, 192.168.100.0/24 with .10 (R1) and .11 (R2)	The simulator works with cables. The addresses are the same as in Phase 1, so the wireless link only changes the medium later.
Router interfaces	One trunk with sub-interfaces per VLAN	One physical interface per active network	The switches are still in VLAN 1. Trunks and Router-on-a-Stick come with the VLANs in week 10.
Switches	VLANs 10, 20, 30 and 99 with trunks	SW-A1, SW-A2 and SW-B1 in VLAN 1	The first goal was to prove addressing, DHCP and routing between sites.
Switch roles	Four switches. In each site SW-x1 receives the trunk from the router and SW-x2 holds the access ports, joined by an EtherChannel of 4 links	Three switches, each one an independent access switch: SW-A1 (Users A), SW-A2 (Servers A) and SW-B1 (Users B). No EtherChannel	The prototype has one network per switch and one cable per router port. SW-B2 is not built because Site B has only users, and the trunk and EtherChannel roles of SW-x1 and SW-x2 come with the VLANs in week 10 and week 11.
Routing	OSPF	Three temporary static routes	Static routes are the simplest way to prove the inter-site path. OSPF replaces them in Phase 3.
NAT and firewall	Planned	Not configured	They need the Internet side and the VLAN policy, which are not built yet.
Addressing plan	VLSM from 10.10.0.0/16	Same plan. 3 subnets active and 5 reserved	No address was changed, so Phase 1 and Phase 2 stay consistent.

Table 1.1. Changes from Phase 1 and their reasons.

1.2 Implemented and planned

Component	Status	Where in this report
Addressing with VLSM, three active subnets	**Implemented**	Section 2
DHCP for Users A (R1) and Users B (R2)	**Implemented**	Section 4
Static addressing for the two servers	**Implemented**	Section 4
Static routes between the sites	**Implemented**	Section 5
SSH version 2 with a local user on R1 and R2	**Implemented**	Section 6
VLANs 10, 20, 30, 99 and 802.1Q trunks	**Planned**	Section 8
Router-on-a-Stick	**Planned**	Section 8
EtherChannel (LACP) and RSTP	**Planned**	Section 8
OSPF	**Planned**	Section 8
NAT and firewall rules	**Planned**	Section 8
OpenWrt on Raspberry Pi	**Planned**	Section 8

Table 1.2. Implemented and planned components.

1.3 Cable connections

From	Port	To	Port	Cable	Network
R1	Gi0/0	SW-A1	Gi0/1	Straight-through	Users A, 10.10.0.0/26
R1	Gi0/1	SW-A2	Gi0/1	Straight-through	Servers A, 10.10.0.128/27
R1	Gi0/2	R2	Gi0/2	Cross-over	Link, 192.168.100.0/24
R2	Gi0/0	SW-B1	Gi0/1	Straight-through	Users B, 10.10.0.64/26
SW-A1	Fa0/1, Fa0/2	PC-A1, PC-A2	Fa0	Straight-through	Users A
SW-A2	Fa0/1, Fa0/2	SRV-Chat, SRV-Auth	Fa0	Straight-through	Servers A
SW-B1	Fa0/1, Fa0/2	PC-B1, PC-B2	Fa0	Straight-through	Users B

Table 1.3. Cable connections of the prototype.

2. Addressing plan with VLSM

The plan is the one of Phase 1: the block 10.10.0.0/16 is split with VLSM, from the largest subnet to the smallest, so no address range is wasted. The link between the routers uses 192.168.100.0/24, outside the block. Table 2.1 lists every subnet. The status column says which ones exist in the prototype.

Subnet purpose	Site	VLAN	Hosts needed	Network / prefix	Mask	Usable range	Broadcast	Gateway	Allocation	Status
Users A	A	10	50	10.10.0.0/26	255.255.255.192	10.10.0.1 - 10.10.0.62	10.10.0.63	10.10.0.1	DHCP, .1 to .10 excluded	**Implemented**
Users B	B	10	50	10.10.0.64/26	255.255.255.192	10.10.0.65 - 10.10.0.126	10.10.0.127	10.10.0.65	DHCP, .65 to .74 excluded	**Implemented**
Servers A	A	20	20	10.10.0.128/27	255.255.255.224	10.10.0.129 - 10.10.0.158	10.10.0.159	10.10.0.129	Static: .130 chat, .131 auth	**Implemented**
Guests A	A	30	20	10.10.0.160/27	255.255.255.224	10.10.0.161 - 10.10.0.190	10.10.0.191	10.10.0.161	DHCP	**Reserved**
Guests B	B	30	20	10.10.0.192/27	255.255.255.224	10.10.0.193 - 10.10.0.222	10.10.0.223	10.10.0.193	DHCP	**Reserved**
Servers B	B	20	10	10.10.0.224/28	255.255.255.240	10.10.0.225 - 10.10.0.238	10.10.0.239	10.10.0.225	Static	**Reserved**
Administration A	A	99	10	10.10.0.240/28	255.255.255.240	10.10.0.241 - 10.10.0.254	10.10.0.255	10.10.0.241	Static	**Reserved**
Administration B	B	99	10	10.10.1.0/28	255.255.255.240	10.10.1.1 - 10.10.1.14	10.10.1.15	10.10.1.1	Static	**Reserved**
Link R1 - R2 (outside the block)	A - B	-	2	192.168.100.0/24	255.255.255.0	192.168.100.1 - 192.168.100.254	192.168.100.255	None (R1 .10, R2 .11)	Static	**Implemented**

Table 2.1. Addressing plan. VLAN numbers are the ones planned for week 10.

- The gateway of each subnet is its first usable address. The servers use fixed addresses: 10.10.0.130 for SRV-Chat and 10.10.0.131 for SRV-Auth.

- The eight subnets of the block end at 10.10.1.15. The rest of 10.10.0.0/16, from 10.10.1.16, is free for growth.

- The DHCP pools exclude the first ten addresses of each user subnet, so the gateways and any fixed device are never leased.

3. Devices, interfaces and switch tables

Table 3.1 lists the devices and their addresses. The router interface descriptions are in Spanish, as configured in the simulator.

Device	Model	Interface	IP address / mask	Description
R1	Cisco 2911	Gi0/0	10.10.0.1 /26	LAN-Usuarios-SitioA
R1	Cisco 2911	Gi0/1	10.10.0.129 /27	LAN-Servidores-SitioA
R1	Cisco 2911	Gi0/2	192.168.100.10 /24	Enlace-a-R2
R2	Cisco 2911	Gi0/0	10.10.0.65 /26	LAN-Usuarios-SitioB
R2	Cisco 2911	Gi0/1	Not used, shutdown	-
R2	Cisco 2911	Gi0/2	192.168.100.11 /24	Enlace-a-R1
SW-A1, SW-A2, SW-B1	Catalyst 2960-24TT	Fa0/1, Fa0/2, Gig0/1	No IP address	Layer 2 only, VLAN 1
PC-A1, PC-A2	PC	Fa0	DHCP, pool USERS-A	Users A
PC-B1, PC-B2	PC	Fa0	DHCP, pool USERS-B	Users B
SRV-Chat	Server	Fa0	10.10.0.130 /27, gateway 10.10.0.129	Chat server (Go)
SRV-Auth	Server	Fa0	10.10.0.131 /27, gateway 10.10.0.129	Authentication (FastAPI)

Table 3.1. Devices and interfaces.

3.1 Interface status

Figure 4 and Figure 5 show `show ip interface brief` on both routers. The interfaces in use have the planned addresses.

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.0.1       YES manual up                    up
GigabitEthernet0/1     10.10.0.129     YES manual up                    up
GigabitEthernet0/2     192.168.100.10  YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Figure 4. *R1 interfaces. Gi0/2 appears up/down because this capture was taken before R2 was configured. After R2 came up, the link worked as shown by the tests.*

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.0.65      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     192.168.100.11  YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Figure 5. *R2 interfaces. Gi0/1 is shut down because it is not used in this prototype.*

3.2 MAC address tables

Each switch learned the MAC addresses of the devices connected to it. On SW-A1 (Figure 6) the two PCs appear on Fa0/1 and Fa0/2, and the router appears on Gig0/1, the port toward R1. SW-A2 and SW-B1 show the same behavior for their own devices (Figure 7 and Figure 8). The tables were read after a ping, because the entries age out when the switch has no traffic.

```
SW-A1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.437c.b497    DYNAMIC     Fa0/2
   1    0001.6340.3538    DYNAMIC     Fa0/1
   1    0090.0c67.4401    DYNAMIC     Gig0/1
```

Figure 6. *SW-A1 MAC address table.*

```
SW-A2#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0090.0c67.4402    DYNAMIC     Gig0/1
   1    00d0.970a.d222    DYNAMIC     Fa0/1
   1    00e0.a392.d919    DYNAMIC     Fa0/2
```

Figure 7. *SW-A2 MAC address table.*

```
SW-B1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0007.ec28.2729    DYNAMIC     Fa0/1
   1    0040.0b0c.4701    DYNAMIC     Gig0/1
   1    00e0.a35c.1864    DYNAMIC     Fa0/2
```

Figure 8. *SW-B1 MAC address table.*

4. DHCP and static server addressing

R1 and R2 are the DHCP servers of their own site. Each pool covers the whole user subnet and excludes the first ten addresses. The servers do not use DHCP.

Pool	Router	Network	Default router	Excluded	First lease
USERS-A	R1	10.10.0.0 /26	10.10.0.1	10.10.0.1 - 10.10.0.10	10.10.0.11
USERS-B	R2	10.10.0.64 /26	10.10.0.65	10.10.0.65 - 10.10.0.74	10.10.0.75

Table 4.1. DHCP pools.

Figure 9 and Figure 10 show the pool configuration on each router.

```
R1#show running-config | section dhcp
ip dhcp excluded-address 10.10.0.1 10.10.0.10
ip dhcp pool USERS-A
 network 10.10.0.0 255.255.255.192
 default-router 10.10.0.1
```

Figure 9. *DHCP configuration on R1.*

```
R2#show running-config | section dhcp
ip dhcp excluded-address 10.10.0.65 10.10.0.74
ip dhcp pool USERS-B
 network 10.10.0.64 255.255.255.192
 default-router 10.10.0.65
```

Figure 10. *DHCP configuration on R2.*

|Client	Site	Address	Mask	Gateway	Obtained from
PC-A1	A	10.10.0.11	255.255.255.192	10.10.0.1	R1, USERS-A
PC-A2	A	10.10.0.12	255.255.255.192	10.10.0.1	R1, USERS-A
PC-B1	B	10.10.0.75	255.255.255.192	10.10.0.65	R2, USERS-B
PC-B2	B	10.10.0.76	255.255.255.192	10.10.0.65	R2, USERS-B
SRV-Chat	A	10.10.0.130	255.255.255.224	10.10.0.129	Static
SRV-Auth	A	10.10.0.131	255.255.255.224	10.10.0.129	Static

Table 4.2. Addresses received by the clients.

The lease table of R1 (`show ip dhcp binding`) lists the two clients of Site A:

IP address	Client hardware address	Type
10.10.0.11	0001.6340.3538	Automatic
10.10.0.12	0001.437C.B497	Automatic

Table 4.3. DHCP leases on R1, taken from the console output of `show ip dhcp binding`.

The IP configuration of each client is in Figure 11 to Figure 14, and the servers in Figure 15 and Figure 16.



Figure 11. *PC-A1 receives 10.10.0.11.*



Figure 12. *PC-A2 receives 10.10.0.12.*



Figure 13. *PC-B1 receives 10.10.0.75.*



Figure 14. *PC-B2 receives 10.10.0.76.*

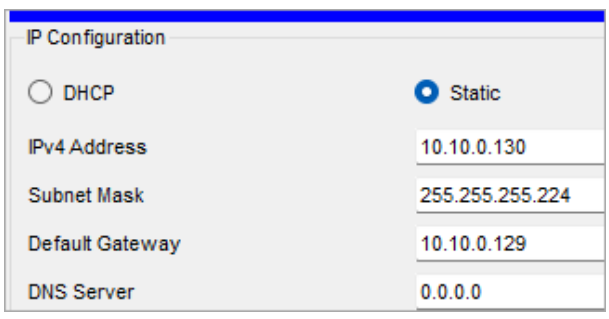


Figure 15. *SRV-Chat, static 10.10.0.130 /27.*

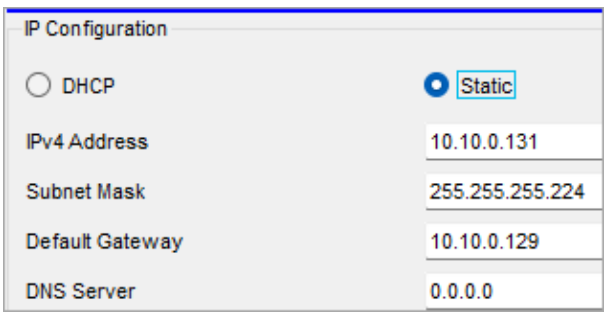


Figure 16. *SRV-Auth, static 10.10.0.131 /27.*

5. Static routing between sites

Each router knows only its own networks. To reach the other site it needs a route through the link 192.168.100.0/24. Three static routes were added. They are temporary: in Phase 3 OSPF replaces them.

Router	Destination	Mask	Next hop	Purpose
R1	10.10.0.64	255.255.255.192	192.168.100.11	Users B
R2	10.10.0.0	255.255.255.192	192.168.100.10	Users A
R2	10.10.0.128	255.255.255.224	192.168.100.10	Servers A

Table 5.1. Static routes.

```
R1#show ip route static
     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S       10.10.0.64/26 [1/0] via 192.168.100.11
```

Figure 17. *Static route on R1 (`show ip route`).*

```
R2#show ip route static
     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
S       10.10.0.0/26 [1/0] via 192.168.100.10
S       10.10.0.128/27 [1/0] via 192.168.100.10
```

Figure 18. *Static routes on R2 (`show ip route`).*

With the routes in place, `tracert 10.10.0.75` from PC-A1 shows the path (Figure 19): the first hop is the gateway R1 (10.10.0.1), the second is R2 on the link (192.168.100.11) and the third is PC-B1. A ping across the sites returns TTL 126, which is 128 minus the two routers.

```
C:\>tracert 10.10.0.75

Tracing route to 10.10.0.75 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.10.0.1
  2    0 ms      0 ms      0 ms      192.168.100.11
  3    0 ms      0 ms      0 ms      10.10.0.75

Trace complete.
```

Figure 19. *Path from PC-A1 to PC-B1.*

A note on the first ping: the first request of each new destination times out because the router has to resolve the ARP entry. The next requests succeed. This is normal behavior and is recorded in the matrix.

6. Remote access and problems found

6.1 SSH on the routers

Remote management uses SSH version 2 with a local user. Telnet is not allowed. Passwords are stored encrypted (`service password-encryption`, `enable secret`). The password of the user is not shown in this report.

Item	Configuration
Domain and version	`ip domain-name uide.local`, `ip ssh version 2`
Keys	RSA, 1024 bits, generated on each router
User	`username admin secret <oculto>`
Lines	`line vty 0 4`, `login local`, `transport input ssh`
Name resolution	`no ip domain-lookup`

Table 6.1. SSH configuration.

Figure 20 shows R1 before the keys were generated (SSH disabled) and Figure 21 and Figure 22 show SSH enabled on R1 and R2. The remote logins are tests T-11 and T-12 (Figure 23 and Figure 24). The password is typed in the prompt and the captures do not show it.

```
R1#show ip ssh
SSH Disabled - version 2
%Please create RSA keys (of atleast 768 bits size) to enable SSH v2.
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

Figure 20. *R1 before the RSA keys: SSH Disabled.*

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

Figure 21. *R1 with SSH version 2 enabled.*

```
R2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R2#
```

Figure 22. *R2 with SSH version 2 enabled.*

```
C:\>ssh -l admin 10.10.0.1

Password:



R1>exit
```

Figure 23. *T-11: PC-A1 logs in to R1 by SSH.*

```
C:\>ssh -l admin 10.10.0.65

Password:



R2>
```

Figure 24. *T-12: PC-B1 logs in to R2 by SSH.*

6.2 Problems found and how they were solved

Problem 1. No route between the sites. The first ping from PC-A1 to PC-B1 failed.

Symptom	Destination host unreachable from 10.10.0.1, 0 of 4 replies (Figure 25).
Cause	`show ip route` on R1 had no route to 10.10.0.64/26 (Figure 26).
Solution	Three static routes (Table 5.1, Figure 17 and Figure 18).
Result	The same ping works with TTL 126 (Figure 27) and the path is confirmed in Figure 19.

```
C:\>ping 10.10.0.75

Pinging 10.10.0.75 with 32 bytes of data:

Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.
Reply from 10.10.0.1: Destination host unreachable.

Ping statistics for 10.10.0.75:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 25. *T-07: ping before the routes (fails).*

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.10.0.0/26 is directly connected, GigabitEthernet0/0
L       10.10.0.1/32 is directly connected, GigabitEthernet0/0
C       10.10.0.128/27 is directly connected, GigabitEthernet0/1
L       10.10.0.129/32 is directly connected, GigabitEthernet0/1
     192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/2
L       192.168.100.10/32 is directly connected, GigabitEthernet0/2
```

Figure 26. *R1 routing table before the routes.*

```
C:\>ping 10.10.0.75

Pinging 10.10.0.75 with 32 bytes of data:

Request timed out.
Reply from 10.10.0.75: bytes=32 time<1ms TTL=126
Reply from 10.10.0.75: bytes=32 time<1ms TTL=126
Reply from 10.10.0.75: bytes=32 time<1ms TTL=126

Ping statistics for 10.10.0.75:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 27. *T-07b: the same ping after the routes (works).*

Problem 2. SSH was disabled. The routers had no RSA keys.

Symptom	`show ip ssh` said SSH Disabled (Figure 20).
Cause	SSH needs an RSA key pair. The key had not been generated.
Solution	The form with the modulus in the same line was rejected by this Packet Tracer version. The key was generated with the interactive command `crypto key generate rsa` and 1024 as the modulus.
Result	SSH version 2 is enabled on both routers (Figure 21 and Figure 22) and the logins work.

Problem 3. A replaced PC received the wrong address. After PC-A1 was deleted and added again, it received 10.10.0.13 instead of 10.10.0.11.

Cause	R1 kept the old lease of the previous PC, so the pool gave the next free address.
Solution	The command to clear all bindings is not accepted in this version. The pool USERS-A was removed and created again with the same parameters.
Result	PC-A1 received 10.10.0.11 and PC-A2 10.10.0.12, as listed in Table 4.3 (Figure 11 and Figure 12).

7. Test matrix

The matrix has 15 tests with their evidence. Each row says what was done, what was expected and what was observed. The figure numbers point to the screenshots at the end of this section. The configurations of both routers and the three switches are saved with copy running-config startup-config.

ID	Source	Destination	Procedure	Expected result	Observed result	Evidence	Status
T-01	PC-A1	PC-A2 (10.10.0.12)	`ping 10.10.0.12`	Replies inside the same subnet	4 of 4 replies, TTL 128	Fig. 28	**Pass**
T-02	PC-A1	Gateway R1 (10.10.0.1)	`ping 10.10.0.1`	Replies from the gateway	4 of 4 replies, TTL 255	Fig. 28	**Pass**
T-03	PC-A1	SRV-Chat (10.10.0.130)	`ping 10.10.0.130`	Replies, routed by R1	3 of 4 replies. The first request timed out (ARP). TTL 127	Fig. 28	**Pass**
T-04	PC-A1	SRV-Auth (10.10.0.131)	`ping 10.10.0.131`	Replies, routed by R1	3 of 4 replies. The first request timed out (ARP). TTL 127	Fig. 30	**Pass**
T-05	PC-B1	PC-B2 (10.10.0.76)	`ping 10.10.0.76`	Replies inside the same subnet	4 of 4 replies, TTL 128	Fig. 29	**Pass**
T-06	PC-B1	Gateway R2 (10.10.0.65)	`ping 10.10.0.65`	Replies from the gateway	4 of 4 replies, TTL 255	Fig. 29	**Pass**
T-07	PC-A1	PC-B1 (10.10.0.75)	`ping 10.10.0.75` before the static routes	Replies across the sites	Destination host unreachable from 10.10.0.1. 0 of 4 replies	Fig. 25, Fig. 26	**Fail (fixed, see 6.2)**
T-07b	PC-A1	PC-B1 (10.10.0.75)	Same ping after the static routes	Replies across the sites	3 of 4 replies. The first request timed out (ARP). TTL 126	Fig. 27	**Pass**
T-08	PC-A1	PC-B1 (10.10.0.75)	`tracert 10.10.0.75`	Path R1, R2, PC-B1	Hops 10.10.0.1, 192.168.100.11 and 10.10.0.75	Fig. 19	**Pass**
T-09	PC-B1	SRV-Chat (10.10.0.130)	`ping 10.10.0.130`	Replies across the sites	4 of 4 replies, TTL 126	Fig. 31	**Pass**
T-10	PC-B1	SRV-Auth (10.10.0.131)	`ping 10.10.0.131`	Replies across the sites	4 of 4 replies, TTL 126	Fig. 31	**Pass**
T-11	PC-A1	R1 (10.10.0.1) by SSH	`ssh -l admin 10.10.0.1`	Password prompt, then the R1 prompt	Login accepted, prompt R1>	Fig. 23	**Pass**
T-12	PC-B1	R2 (10.10.0.65) by SSH	`ssh -l admin 10.10.0.65`	Password prompt, then the R2 prompt	Login accepted, prompt R2>	Fig. 24	**Pass**
T-13	PC-A1, PC-A2	R1 (pool USERS-A)	Set the PCs to DHCP, then `show ip dhcp binding` on R1	Addresses from 10.10.0.11, mask /26, gateway 10.10.0.1	PC-A1 10.10.0.11 and PC-A2 10.10.0.12, gateway 10.10.0.1. Both leases listed on R1	Fig. 9, Fig. 11, Fig. 12, lease table in section 4	**Pass**

ID	Source	Destination	Procedure	Expected result	Observed result	Evidence	Status
T-14	PC-B1, PC-B2	R2 (pool USERS-B)	Set the PCs to DHCP	Addresses from 10.10.0.75, mask /26, gateway 10.10.0.65	PC-B1 10.10.0.75 and PC-B2 10.10.0.76, gateway 10.10.0.65	Fig. 10, Fig. 13, Fig. 14	Pass

Table 7.1. Preliminary test matrix.

7.1 Pending tests

These tests belong to the parts of the design that are not built yet. They will be run when each part is configured.

ID	Planned test	Requirement	Procedure	Expected result	Status
P-01	VLANs 10, 20, 30 and 99	RF-NET-02	`show vlan brief` on each switch	VLANs exist and the ports are assigned	Pending
P-02	802.1Q trunks	RF-NET-02	`show interfaces trunk`	Trunks carry VLANs 10, 20, 30 and 99	Pending
P-03	Router-on-a-Stick	RF-NET-02	Ping between VLANs through the sub-interfaces	Inter-VLAN routing works as the policy allows	Pending
P-04	EtherChannel	RF-NET-04	`show etherchannel summary`, then disconnect 1 of the 4 cables	The bundle stays up with the remaining links	Pending
P-05	RSTP	RF-NET-04	`show spanning-tree`	Correct root bridge and port roles	Pending
P-06	OSPF	RF-NET-03	`show ip ospf neighbor` and `show ip route` after removing the static routes	R1 and R2 are full neighbors and learn the routes of the other site	Pending
P-07	NAT	RF-SEC-03	Ping to the Internet from each site	Each site reaches the Internet on its own	Pending
P-08	Firewall	RF-SEC-03	Guests to Servers and Users to Administration	Blocked traffic is denied	Pending
P-09	DHCP for Guests	RF-NET-01	Set a guest device to DHCP	Address from the Guests pool	Pending
P-10	Go client from Site B	RF-INT-01	Log in and send a message to the chat server in Site A	Login and message delivery work across the sites	Pending

Table 7.2. Tests planned for the next weeks.

7.2 Evidence

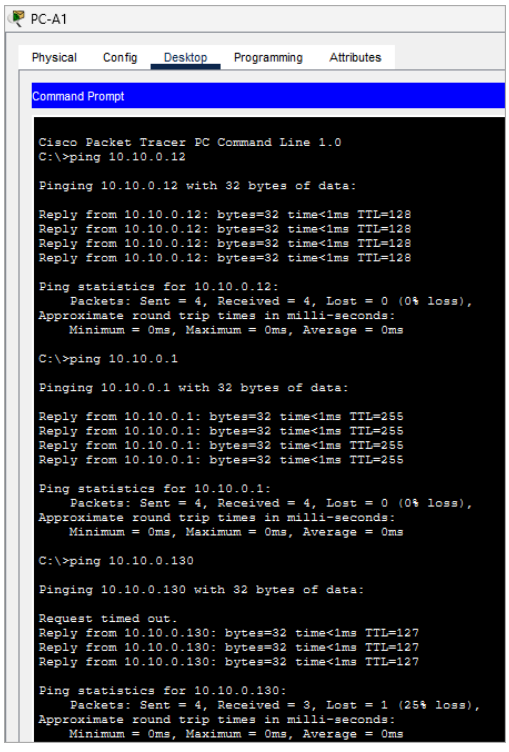


Figure 28. *T-01 to T-03: pings from PC-A1 to PC-A2, R1 and SRV-Chat.*

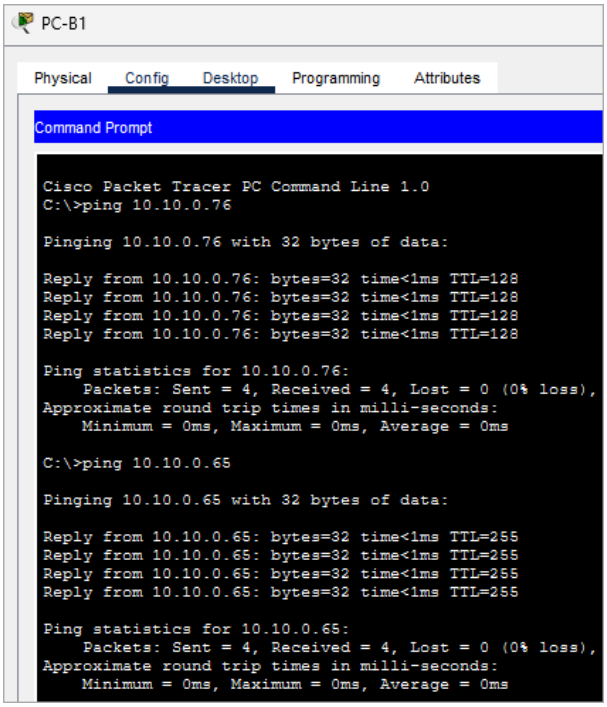


Figure 29. *T-05 and T-06: pings from PC-B1 to PC-B2 and R2.*

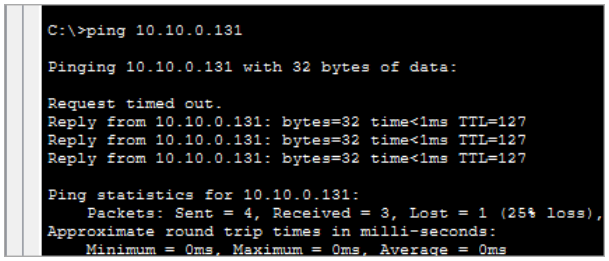


Figure 30. *T-04: ping from PC-A1 to SRV-Auth.*

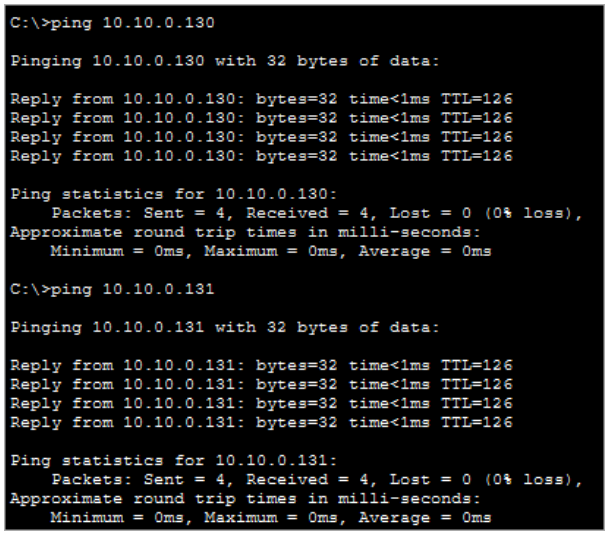


Figure 31. *T-09 and T-10: pings from PC-B1 to SRV-Chat and SRV-Auth.*

8. Remaining work and requirements status

The prototype proves addressing, DHCP, routing between the sites and secure remote access. The parts below are still to be done.

Requirement	Description	Status	Owner
RF-NET-01	Addressing and DHCP with VLSM	**Met in simulation**	Darwin Alcarraz
RF-NET-02	VLANs, trunks and inter-VLAN routing	**Pending**	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
RF-NET-03	Dynamic routing between sites (OSPF)	**Pending**	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
RF-NET-04	Redundancy (EtherChannel and RSTP)	**Pending**	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
RF-SEC-03	NAT and firewall policy	**Pending**	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida

Table 8.1. Status and owner of the network requirements.

8.1 Migration to OpenWrt

The final network uses OpenWrt on Raspberry Pi. Table 8.2 maps each function of the Cisco prototype to its OpenWrt equivalent, so the work done in Packet Tracer is not lost. The Cisco commands are not copied to OpenWrt: only the concepts are translated, and the OpenWrt configuration (UCI, firewall zones and FRRouting) is written for that system.

Function	Cisco prototype	OpenWrt
Interface addresses	`interface` and `ip address`	`/etc/config/network`
DHCP	`ip dhcp pool`	dnsmasq, `/etc/config/dhcp`
Excluded addresses	`ip dhcp excluded-address`	Start and limit of the dnsmasq range
Static routes	`ip route`	`route` sections in `/etc/config/network`
Dynamic routing	OSPF on the router	FRRouting (OSPF)
VLANs	`vlan`, trunks, sub-interfaces	Bridge VLAN filtering, 802.1Q interfaces
NAT	`ip nat`	Firewall zone with masquerade, excluding 10.10.0.0/16
Firewall	Access lists	Firewall zones and rules
Remote access	SSH on the vty lines	Dropbear
Link between sites	Cross-over cable	Wi-Fi link on wlan0

Table 8.2. From Cisco to OpenWrt.

The migration order is:

1. Create the VLANs, trunks and Router-on-a-Stick in Packet Tracer and run P-01 to P-03. **Owners:** Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida.
2. Add EtherChannel and RSTP and run P-04 and P-05. **Owners:** Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida.
3. Replace the static routes with OSPF and run P-06. **Owners:** Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida.
4. Add NAT and the firewall policy and run P-07 and P-08. **Owners:** Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida.
5. Install OpenWrt on the Raspberry Pi, load the same addressing and repeat the tests. **Owners:** Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida.
6. Connect the Go and FastAPI services and run P-10. **Owners:** Darwin Alcarraz, Cayetano Cordova and Jose Almeida. Darwin Alcarraz leads the Go and FastAPI integration.

A Raspberry Pi has a single Ethernet port. For the final network it needs a USB Ethernet adapter or a VLAN-capable switch to reach several networks, and this is part of the equipment check of week 10.

9. Updated Schedule

Week	Activities	Result	Owner
9	Packet Tracer prototype, tests and this report	Phase 2 delivered	Darwin Alcarraz
10	Equipment check. VLANs, trunks and Router-on-a-Stick in the simulation. OpenWrt installation	P-01 to P-03 passed	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
11	Complete the simulation (EtherChannel, RSTP, OSPF, NAT, firewall). First hardware deployment	P-04 to P-09 passed	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
12	Integration of the Go and FastAPI services in the network	P-10 prepared	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
13-14	End-to-end tests and fixes	Final test matrix	Darwin Alcarraz (lead), Cayetano Cordova and Jose Almeida
15	Defense	Project defended	All three
16	Closure: final review of documentation and repository	Deliverables closed	All three

Table 9.1. Schedule through the end of the course.

The servers will be placed in VLAN 20 of Site A in weeks 11 and 12. The three members work on this step together: Darwin Alcarraz leads the Go and FastAPI integration and the network side, and Cayetano Cordova and Jose Almeida bring the other components.

10. Contributions, AI use, resources and deliverables

10.1 Contributions

Student	Contribution
Darwin Alcarraz	Network component: built the Packet Tracer prototype, configured the routers and switches, ran the tests, took the evidence and wrote this report, except for the parts listed for the other two members. Leads the Go and FastAPI integration (step 6 of section 8) and the repository.
Cayetano Cordova	Prepared the explanation of the missing route problem and its solution (section 6.2). Components of the integrative project for the other subjects (programming and security), developed in parallel. Works on the Go and FastAPI integration in step 6 of section 8, together with the other two members.
Jose Almeida	Prepared the explanation of how DHCP works and of the routing between the sites (sections 4 and 5). Components of the integrative project for the other subjects (programming and security), developed in parallel. Works on the Go and FastAPI integration in step 6 of section 8, together with the other two members.

Table 10.1. Contributions. The responsibilities were reorganized after the work plan of Phase 1 (section 6 of the Phase 1 report).

10.2 Use of AI

Claude (Anthropic) was used as a guide and writing assistant. It explained the steps and helped to draft the text of this report. All commands were typed and run by Darwin Alcarraz in Packet Tracer and all screenshots were taken from his simulation. Figures 1 and 3 were drawn with AI help from the topology that was built. The results were checked against the screenshots.

10.3 Resources

- Cisco Packet Tracer 9.0.1.
- Phase 1 report of the project (architecture and addressing plan).
- Integrative project document and assignment of Phase 2.

10.4 Deliverables

Item	Repository path
Repository	https://github.com/cayetano-c/chat-empresarial
Packet Tracer file	`network/Phase2-Prototype-G6.pkt`
Topology diagrams (updated)	`network/diagrams/Physical-TopologyPh2.png` and `network/diagrams/Logical-TopologyPh2.png`
Addressing table	`network/addressing/addressing-tablePh2.csv`
Configuration backups (R1, R2, SW-A1, SW-A2, SW-B1)	`network/configs/`
Test matrix	`network/test-matrixPh2.csv`
Screenshots	`network/evidence/`
This report	`docs/phase2-report.pdf`

Table 10.2. Files delivered.

GitHub Repository Link: https://github.com/cayetano-c/chat-empresarial

The running configuration of each router and switch was saved to the startup configuration with `copy running-config startup-config`, and the backups in `network/configs/` were taken from that saved state.

No password, token or private key is included in this report or in the repository. The configuration backups show `<oculto>` in place of the password hashes, and the screenshots do not show passwords.